

MATLAB: Session 2

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Exercises.

Part A – Linear independence

- A.1** (*Already done last time*) Write a small MATLAB function that takes as input two vectors (as columns) and checks if they are linearly independent, handling possible zeros safely.

Hint: Start by checking whether the two vectors have the same length, i.e. whether they both belong to the same $\mathbb{R}^n$. It could be useful to follow the MATLAB Onramp course to get familiar with functions.

- A.2** Explain why any set of four vectors in $\mathbb{R}^3$ is linearly dependent. Provide a concrete example of four such vectors and exhibit a non-trivial linear relation among them; verify it in MATLAB.

- A.3** Create the vectors

$$\mathbf{u}_1 = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_3 = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}.$$

Assign them to the variables `u1`, `u2` and `u3`.

- A.4** Show that $\mathbf{u}_3 = \frac{3}{2}\mathbf{u}_1 + \mathbf{u}_2$. Conclude that $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is linearly dependent, while $\{\mathbf{u}_1, \mathbf{u}_2\}$ is independent. Verify in MATLAB, using the function created in **A.1**.

- A.5** Create the vectors

$$\mathbf{s}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{s}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{s}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{s}_4 = \begin{bmatrix} 2 \\ 1 \\ 2 \\ 1 \end{bmatrix}.$$

Assign them to the variables `s1`, `s2`, `s3` and `s4`.

- A.6** Reduce $\{\mathbf{s}_1, \mathbf{s}_2, \mathbf{s}_3, \mathbf{s}_4\}$ to a smaller generating set that is independent. State the dimension of the span and one possible basis; confirm relations in MATLAB, using the function created in **A.1**.

Part B – Bases, subspaces and coordinates

- B.1** For each set, decide if it is a vector subspace of the indicated vector space. If so, give a basis and its dimension.

- a) $T = \{(x, y, x + y) \in \mathbb{R}^3 : x, y \in \mathbb{R}\}$.
- b) $U = \{(x, y, 0) \in \mathbb{R}^3 : x, y \in \mathbb{R}\}$.
- c) $V = \{(t^2, 2t, -t, 0) \in \mathbb{R}^4 : t \in \mathbb{R}\}$.
- d) $W = \{(a, b, a - b, 1) \in \mathbb{R}^4 : a, b \in \mathbb{R}\}$.
- e) $Z = \{(a, b, a - b, c) \in \mathbb{R}^4 : a, b, c \in \mathbb{R}\}$.

- B.2** Verify in MATLAB that the basis of Z found in **B.1** is made of linearly independent vectors.

- B.3** Create the vectors

$$\mathbf{b}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} 0 \\ 3/2 \\ 3/2 \end{bmatrix}, \quad \mathbf{b}_3 = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}.$$

Assign them to the variables **b1**, **b2** and **b3**.

- B.4** Check if $\mathbf{b}_1$, $\mathbf{b}_2$ and $\mathbf{b}_3$ are elements of T . Is $\{\mathbf{b}_1, \mathbf{b}_2\}$ a basis of T ? If yes, find the coordinates of $\mathbf{b}_3$ with respect to the basis $\{\mathbf{b}_1, \mathbf{b}_2\}$. Verify your answers in MATLAB.

- B.5** Find a vector $\mathbf{b}_4$ such that $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_4\}$ is a basis of $\mathbb{R}^3$. Write any vector of $\mathbb{R}^3$ as a linear combination of $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_4\}$. Verify your answers in MATLAB.

- B.6** Create the vectors

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 0 \\ -2 \\ 7 \end{bmatrix}, \quad \mathbf{a}_3 = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}, \quad \mathbf{a}_4 = \begin{bmatrix} 1 \\ 6 \\ -10 \end{bmatrix}, \quad \mathbf{a}_5 = \begin{bmatrix} 3 \\ 6 \\ 12 \end{bmatrix}.$$

Assign them to the variables **a1**, **a2**, **a3**, **a4** and **a5**.

- B.7** Without doing any computations, decide whether these vectors are linearly independent.
- B.8** Check on MATLAB whether the following subsets are made of linearly independent vectors

- $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$;
- $\{\mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4\}$;
- $\{\mathbf{a}_3, \mathbf{a}_4, \mathbf{a}_5\}$;
- $\{\mathbf{a}_1, \mathbf{a}_3, \mathbf{a}_5\}$;
- $\{\mathbf{a}_3, \mathbf{a}_5\}$;
- $\{\mathbf{a}_1, \mathbf{a}_4, \mathbf{a}_5\}$.

Part C – Matrix Multiplication

C.1 Write a MATLAB function that takes as input two matrices A and B , and outputs their row-by-column product $C = AB$. You are not allowed to use the MATLAB multiplication operator `*`, although you may use it afterward to check whether your result is correct.

Warning: Do not forget to check whether the matrices are multiplicatively compatible!

C.2 Create the matrices

$$A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & 2 & 5 \\ 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 7 & 0 & 2 \\ 1 & 0 & 0 & 0 \end{bmatrix}.$$

Assign them to the variables **A** and **B**.

C.3 Compute AB and BA .

C.4 Check whether $AB = BA$.

C.5 Create the matrices

$$E = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \quad F = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 0 \end{bmatrix}, \quad G = \begin{bmatrix} 0 & 1 \\ -1 & 0 \\ 1 & -2 \end{bmatrix}.$$

Assign them to the variables **E**, **F**, **G**.

C.6 For each of the following, first decide (by checking matrix sizes) whether the product is defined; if it is, compute it in MATLAB:

$$EF, \quad EG, \quad FE, \quad FG, \quad (E^\top)F, \quad (G^\top)E.$$

Part D – Symbolic computations

D.1 Create a symbolic variable **x**. Define the expression

$$exp = x^4 + 2x^3 - 15x^2 + 6x^3 + 12x^2 - 90x - 7x^2 - 14x + 105.$$

Assign it to the variable **exp**. Substitute $x = 2$ into **exp** and save the result in **exp2**.

D.2 Factorize *exp* using the command **factor**, saving the result in the variable **fatt**. Simplify *exp* using **simplify**, saving the result in the variable **simpl**. Observe the difference between the two outputs.

D.3 Define the equation

$$x^4 + 8x^3 - 10x^2 - 104x + 105 = 0.$$

Save it in the variable **eq**. Solve **eq** with respect to **x** and save the result in the variable **solx**. Comment on the form of the output.

D.4 Define the symbolic variables **a**, **b**, **c**. Define the expressions

$$eqn_1 = 5a + b + 2c, \quad eqn_2 = 7b - 3c.$$

Assign them to the variables **eqn1** and **eqn2** respectively.

D.5 Define the equation

$$eqn_2 = eqn_1.$$

Save it in the variable **eqn3**.

D.6 Solve **eqn3** with respect to **a** and store the solution in the variable **sol1**. Then substitute $a \mapsto 4a + 2$ directly into **eqn3** and solve again with respect to **a**, saving the new solution in the variable **sol2**.

D.7 Consider the matrix

$$C = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$$

with $a, b, c \in \mathbb{R}$ all distinct. Determine all matrices $D \in \mathbb{R}^{3 \times 3}$ that commute with C , i.e. such that $CD = DC$.